ELECTRICAL SUPPLY UNIT (ESU)

id094000003100

Purpose

- The electrical supply unit (ESU) performs control of the headlights, rear fog light, map light, door lock actuators, rear wiper, front washer and rear washer based on the input signal from the BCM.

Function

- The control content of the electrical supply unit (ESU) is as follows.

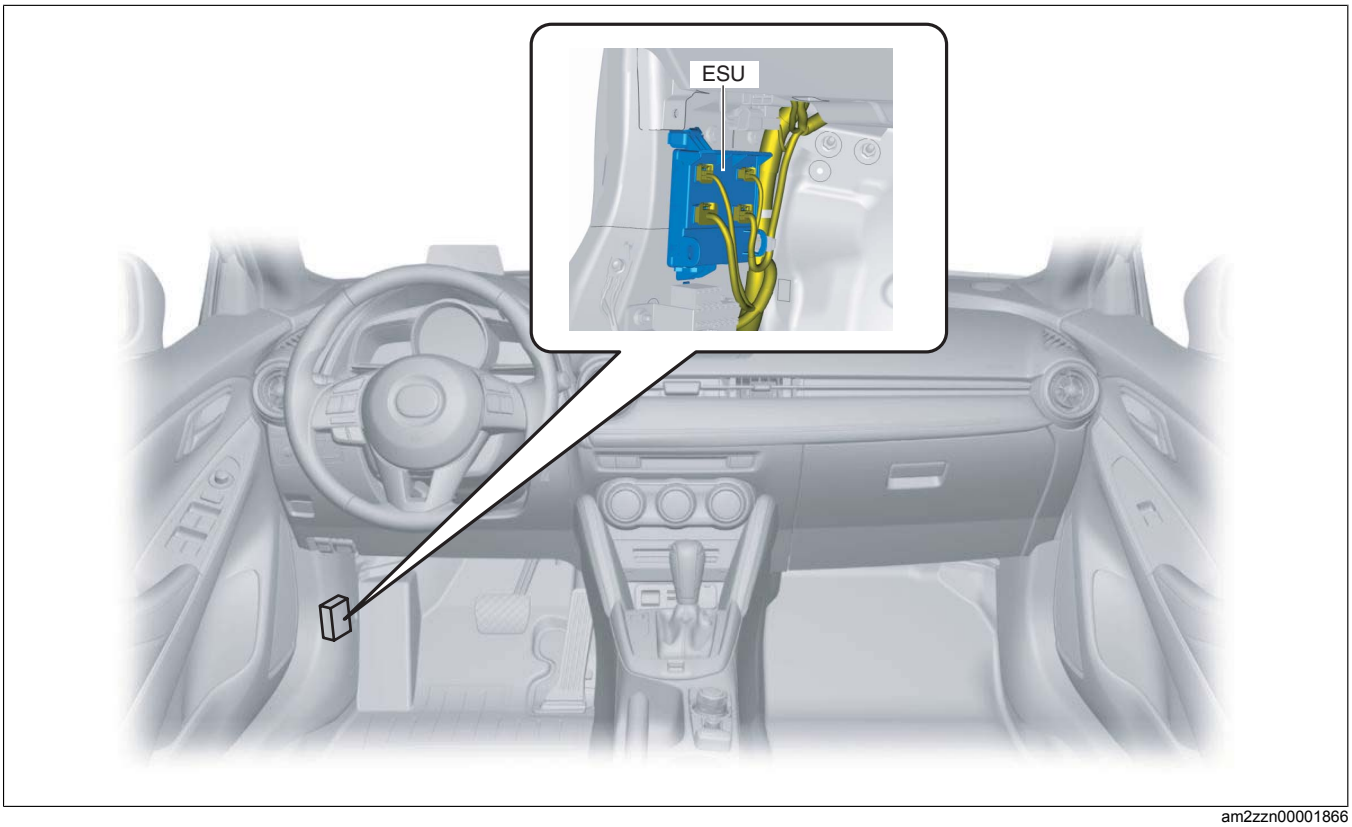
Control table

Control	Content	Reference
Exterior lighting	The ESU illuminates/turns off the headlights based on the signal from the BCM.	(See EXTERIOR LIGHTING SYSTEM.)
Front map light	The ESU illuminates/turns off the front map light based on the signal from the BCM.	(See FRONT MAP LIGHT.)
Rear map light	The ESU illuminates/turns off the rear map light based on the signal from the BCM.	(See REAR MAP LIGHT.)
Cargo compartment light	The ESU illuminates/turns off the cargo compartment light based on the signal from the BCM.	(See CARGO COMPARTMENT LIGHT.)
Power door lock	The ESU operates the door lock actuators based on the signal from the BCM.	(See POWER DOOR LOCK SYSTEM.)
Liftgate opener	The ESU operates the liftgate opener actuator based on the signal from the BCM.	(See LIFTGATE OPENER SYSTEM.)
Rear wiper	The ESU operates the rear wiper motor based on the signal from the BCM.	(See REAR WIPER/WASHER SYSTEM.)
Windshield/rear washer	The ESU operates the washer motor based on the signal from the BCM.	(See WINDSHIELD WIPER/WASHER SYSTEM.) (See REAR WIPER/WASHER SYSTEM.)

Structural View

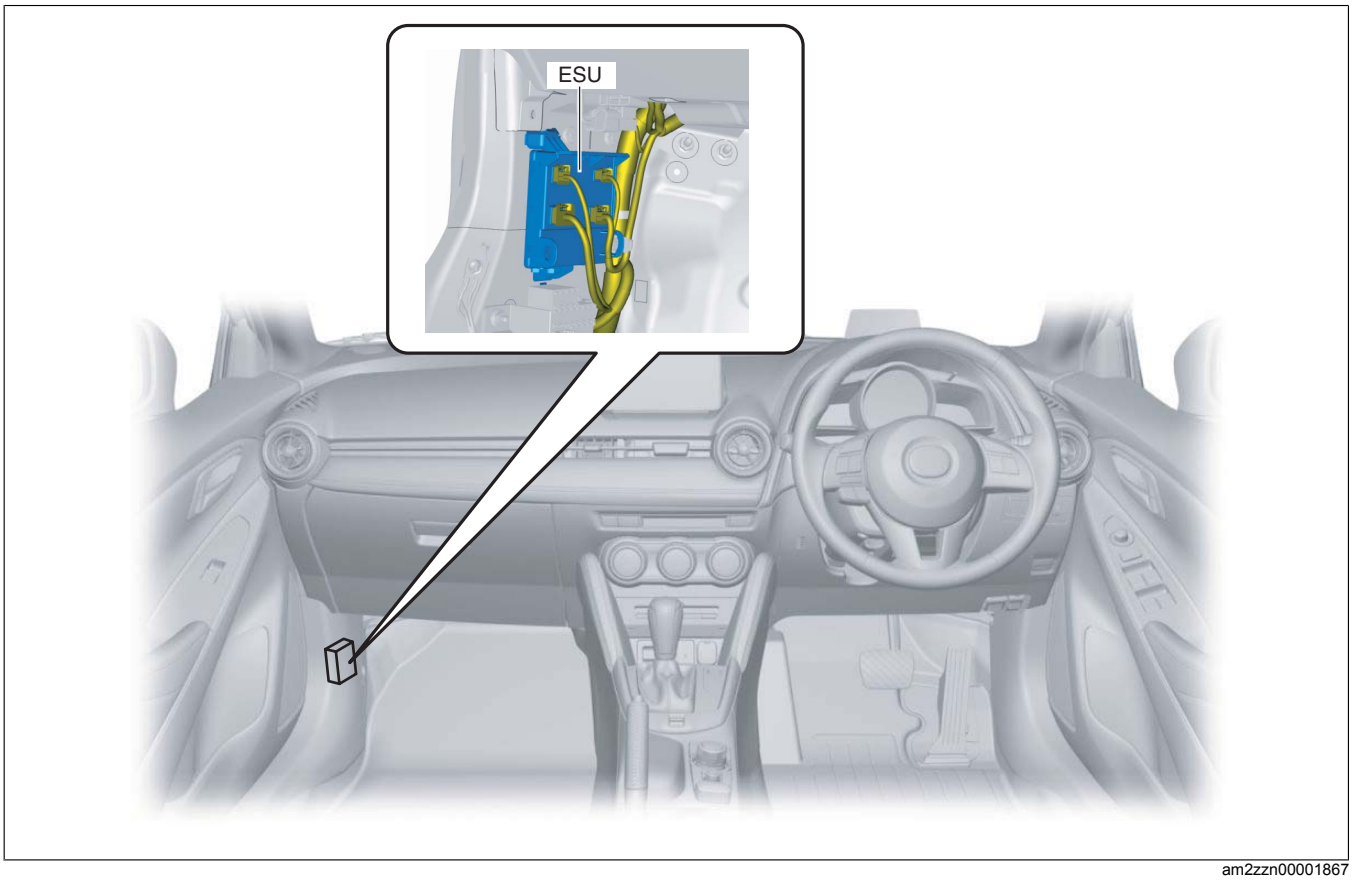
- The electrical supply unit (ESU) is located on the inside of the front side trim.

L.H.D.



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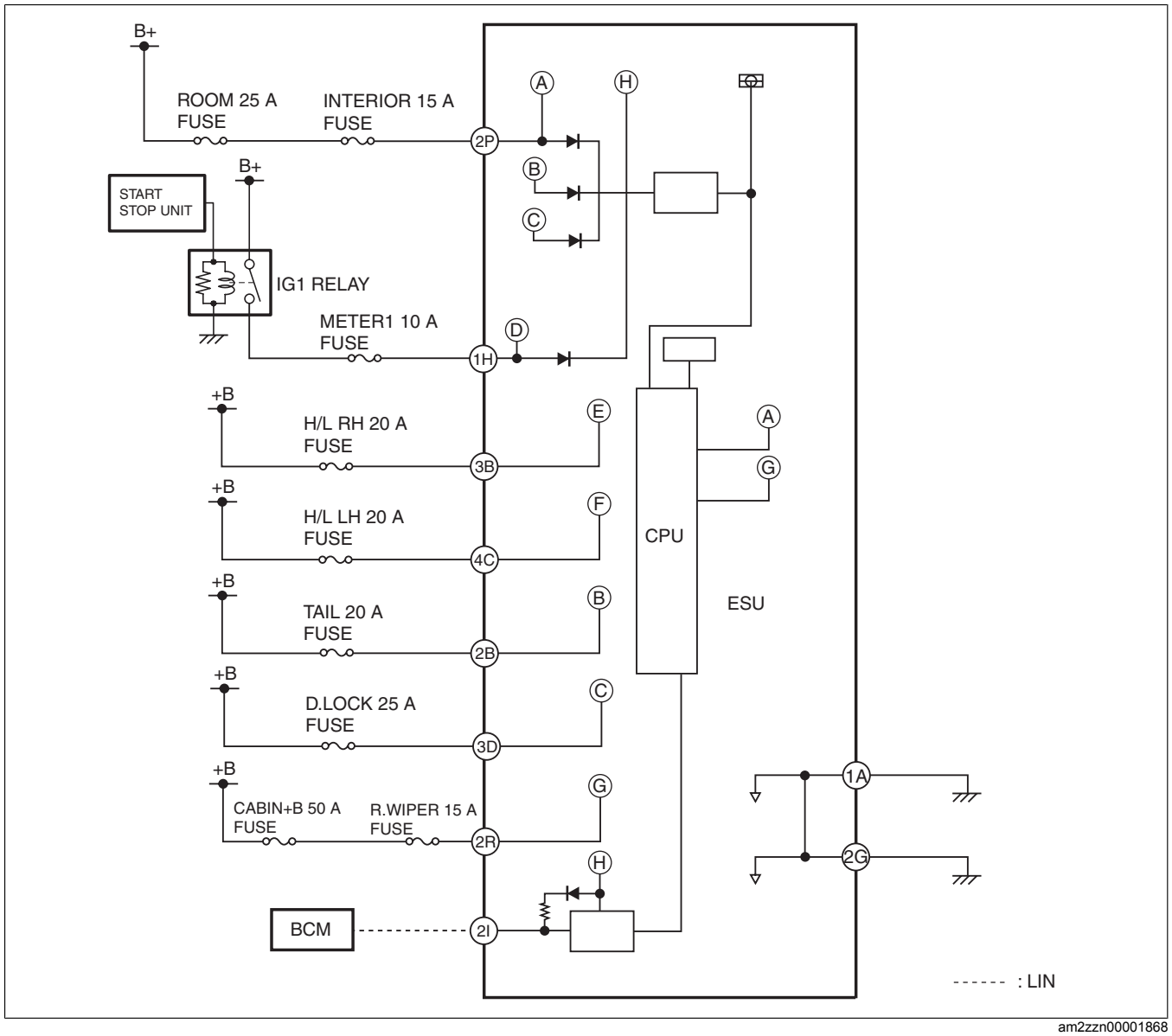
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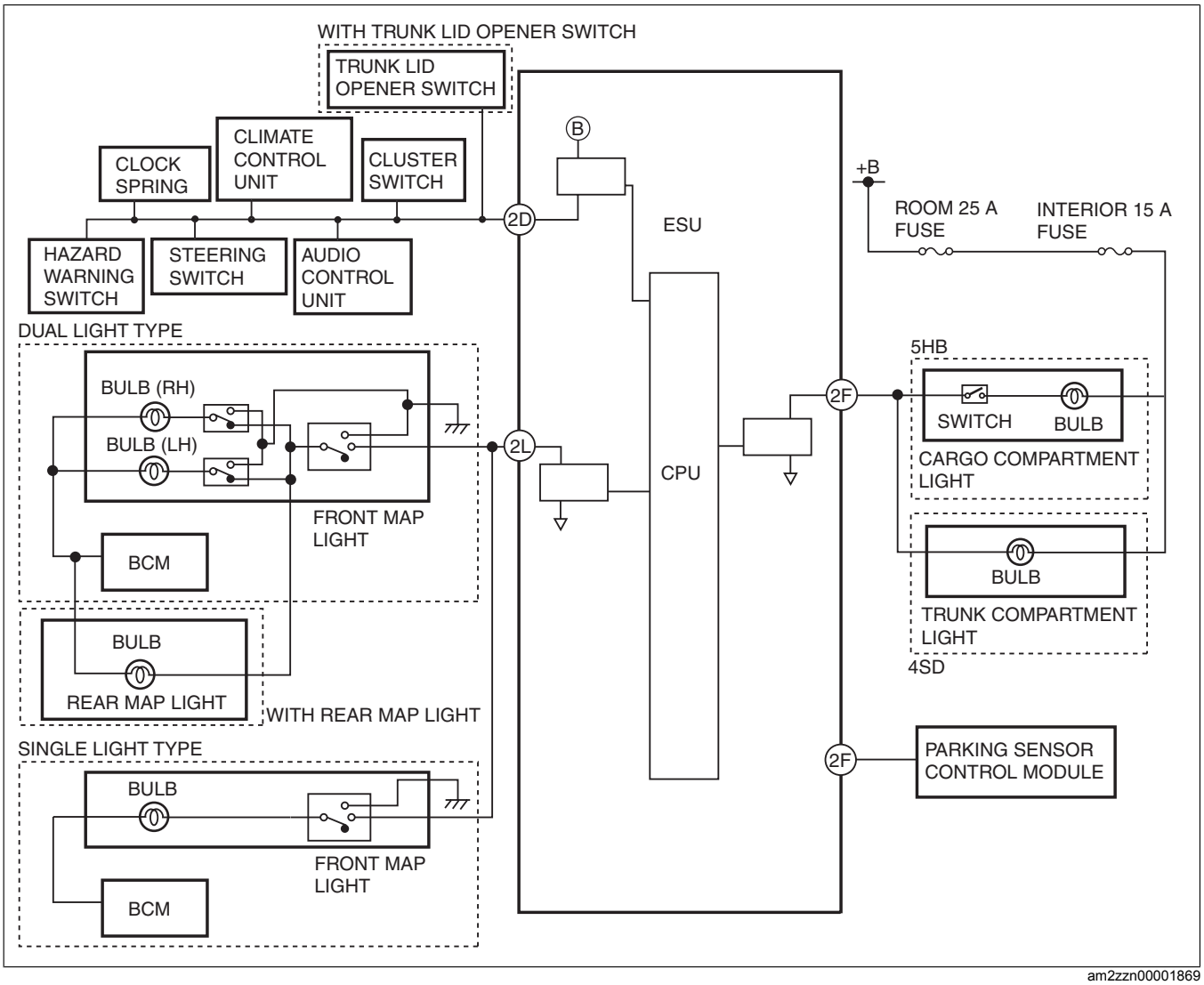
System wiring diagram

Power supply, earth, LIN communication



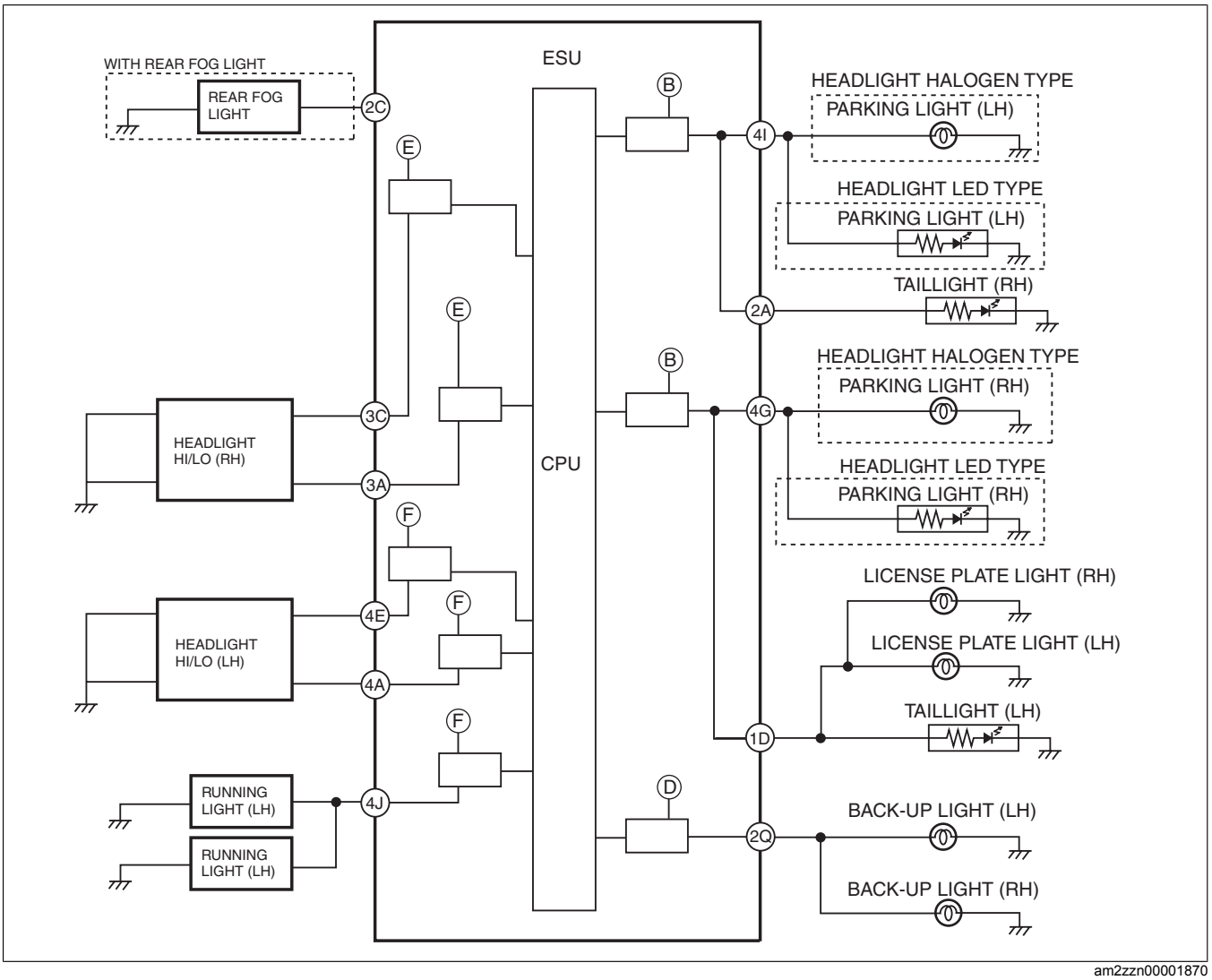
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Map light, cargo compartment light



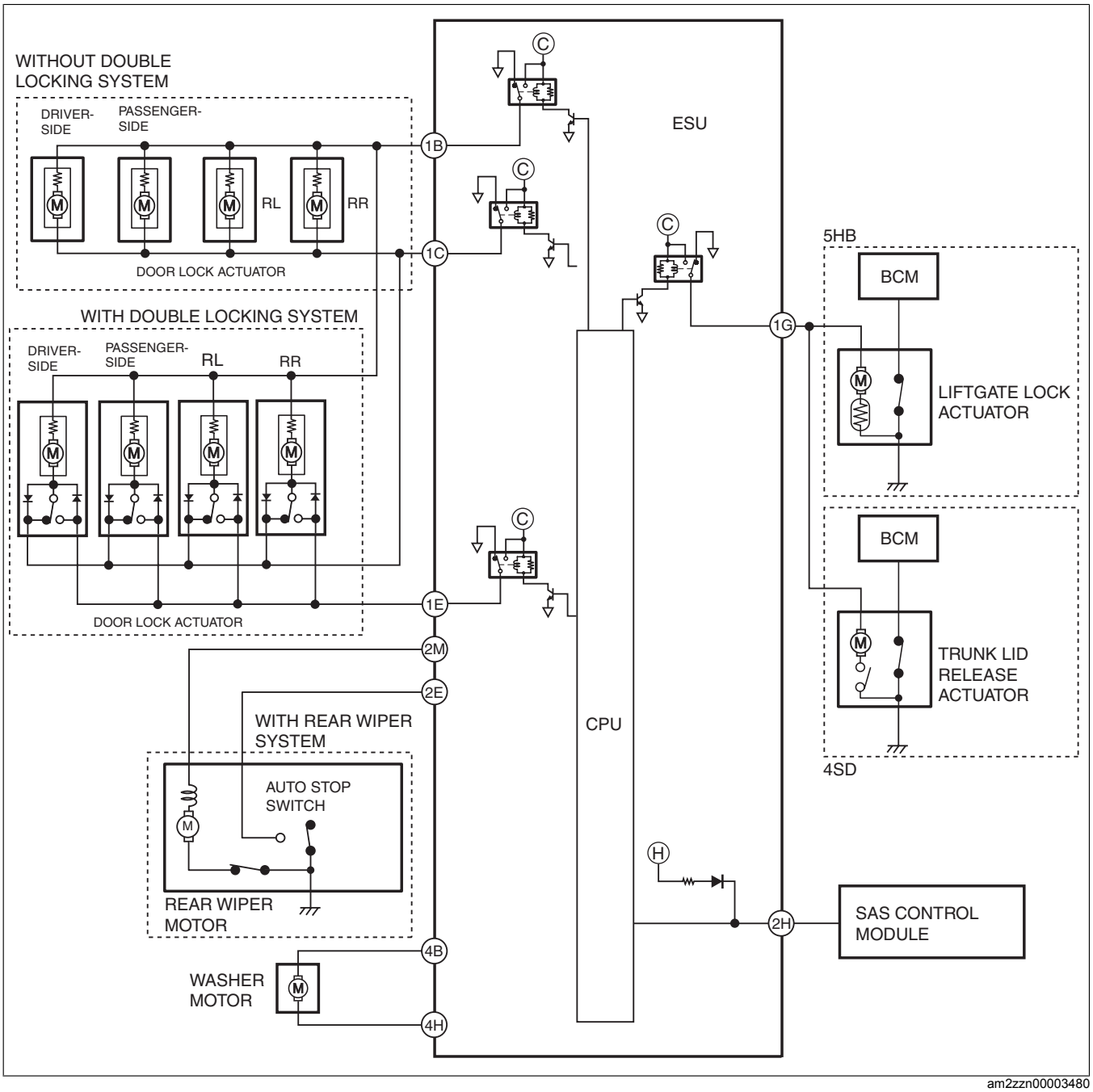
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Exterior lighting



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Door lock, rear wiper



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Operation

For the operation of each control, refer to the following system operations.

- Exterior lighting system control (See EXTERIOR LIGHTING SYSTEM.)
- Front map light control (See FRONT MAP LIGHT.)
- Rear map light control (See REAR MAP LIGHT.)
- Cargo compartment light control (See CARGO COMPARTMENT LIGHT.)
- Power door lock system control (See POWER DOOR LOCK SYSTEM.)
- Liftgate opener system control (See LIFTGATE OPENER SYSTEM.)
- Windshield washer control (See WINDSHIELD WIPER/WASHER SYSTEM.)
- Rear wiper/washer system control (See REAR WIPER/WASHER SYSTEM.)

Fail-safe

Exterior lighting system

- If any of the following malfunctions occurs, the electrical supply unit (ESU) turns the TNS and headlights LO on, and stops other exterior light control.
  - — Malfunction in CPU in electrical supply unit (ESU) detected
  - — LIN communication error (cannot communicate with BCM, receives malfunction signal from BCM)

Room light control system

- If any of the following malfunctions occurs, the electrical supply unit (ESU) turns the room light off.
  - LIN communication error (cannot communicate with BCM, receives malfunction signal from BCM)

Power door lock system

- Not applicable

Liftgate opener system

- Not applicable

Rear wiper system

- If any of the following malfunctions occurs, the electrical supply unit (ESU) stops the rear wiper.
  - Malfunction in CPU in electrical supply unit (ESU) detected
  - LIN communication error (cannot communicate with BCM, receives malfunction signal from BCM)
  - Auto stop switch malfunction (wiper operation status unknown) while wiper is operating is detected

Front/rear washer system

- When the electrical supply unit (ESU) receives a windshield washer operation request signal from the BCM for a continuous **1 min or more**, it stops the front/rear washer control.